

High-Level Architecture of AI Log Analyzer

1. System Overview

The AI Log Analyzer is a domain-agnostic, multilingual, intent-driven system designed to analyze network and system logs across different organizational environments such as universities, hospitals, stations, and enterprises. The system operates in both **Online (continuous)** and **Offline (batch)** modes using a common multi-agent architecture consisting of:

- Intent Agent
- Data Agent
- Analysis Agent
- Report Agent

Natural language prompts written in any language define *what to analyze*. These prompts are interpreted, structured, and applied to log data through chained intelligent agents.

2. Online System Architecture (Continuous Analysis)

Workflow:

1. Preconfigured/User Prompts (NLP)

Monitoring objectives are defined using natural language prompts (any language).

2. Intent Agent

The prompt is interpreted and converted into a structured intent specification.

3. Data Agent

At fixed time intervals, continuous logs from live servers are filtered using the structured intent.

4. Analysis Agent (Single/Multiple)

The extracted data is analyzed using rule-based or ML-based methods, either via a single agent or multiple parallel agents.

5. Report Agent

A standardized report is generated classifying the system state as Good/Bad (or Warning).

Key Property:

Online processing runs continuously at predefined intervals, enabling near real-time system intelligence.

3. Offline System Architecture (One-Time File-Based Analysis)

Workflow:

1. Preloaded Log Files

Historical or exported logs are uploaded to the system.

2. User Prompt (NLP)

The user specifies the type of analysis to be performed.

3. Intent Agent

The prompt is converted into a structured intent.

4. Data Agent

Relevant log fields and records matching the intent are extracted from files.

5. Analysis Agent (Single/Multiple)

Analytical processing is executed on the extracted dataset.

6. Report Agent

A standardized one-time output report is generated.

Key Property:

Offline analysis is executed once per dataset or file and is not time-interval based.

4. Individual Agent Architecture and Responsibilities

4.1 Intent Agent

Role:

Transforms multilingual natural-language user prompts into structured machine-readable intent objects.

Inputs:

- User Prompt (any language)
- Organizational context (optional)

Core Tasks:

- Multilingual NLP understanding
- Goal extraction (security, performance, usage, audit, etc.)
- Entity identification (IP, user, service, system)
- Condition extraction (thresholds, patterns, error codes)

✓ Classful Division of Intent (Newly Added)

After understanding the prompt, the Intent Agent categorizes each intent into predefined **Intent Classes**. This class-based routing determines how the logs will be processed further by the system.

Example Intent Classes:

- **Security Intent** → intrusion detection, brute-force attempts, data exfiltration
- **Performance Intent** → latency, CPU/memory usage, bandwidth spikes
- **Availability Intent** → downtime, service crashes
- **Compliance Intent** → policy violations, access control violations
- **Usage/Analytics Intent** → traffic volume, user behavior

Each class activates a specific analysis pipeline and specialized analysis agents.

Output:

- Structured **Intent Object**
 - Associated **Intent Class Label** for pipeline selection
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4.2 Data Agent

Role:

Filters and extracts only relevant log data required to satisfy the classified intent.

Inputs:

- Intent Object + Intent Class
- Live logs (online) or files (offline)

Core Tasks:

- Parse heterogeneous log formats
- Select relevant log entries using:
 - Time window
 - IP/User filters
 - Service/Event types
- Field extraction and normalization
- Metric pre-computation (counts, averages, trends)

Output:

- Clean structured dataset aligned with the user intent
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4.3 Analysis Agent

Role:

Analyzes extracted data using rule-based logic or ML-based models to determine system state.

Inputs:

- Structured Intent Object
- Extracted Dataset
- Intent Class Label

Core Tasks:

- Apply domain-specific analysis logic
- Support parallel analysis pipelines
- Perform detection such as:
 - Anomaly detection
 - Trend analysis
 - Pattern matching
 - Correlation analysis
- Assign severity, confidence score, or anomaly risk

Output:

- Structured **Analysis Result Object**
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4.4 Report Agent

Role:

Produces standardized human-readable result outputs.

Inputs:

- Analysis Result Object(s)

Core Tasks:

- Convert technical conclusions into:
 - Good / Bad / Warning
- Highlight core issues and impacted resources
- Maintain a consistent reporting format

Output:

- Final AI Log Analysis Report
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5. Online vs Offline System – Core Difference

Feature	Online System	Offline System
Input	Continuous logs	Static files
Processing	Periodic, continuous	One-time
Output	Repeated reports	Single report
Use Case	Real-time monitoring	Audit & forensic analysis

6. Key Architectural Strengths

- Fully domain-agnostic
 - Multilingual prompt support
 - Intent-driven intelligent routing
 - Class-based intent orchestration
 - Modular agent architecture
 - Scalable to real-time and batch workflows
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